



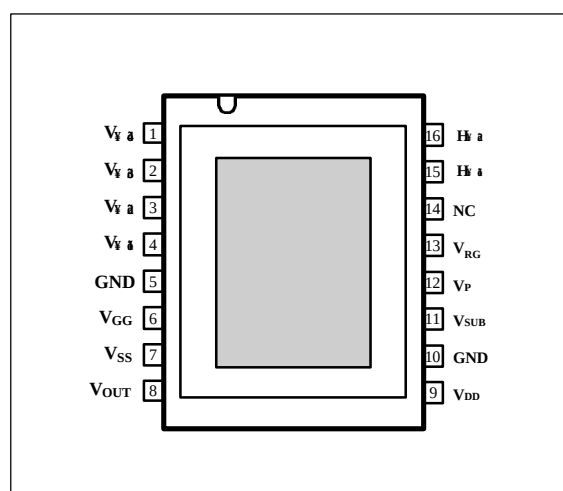
Description

The Ai329DB is a 290K pixels CCD area sensor for CCIR 1/3inch video cameras. Buried photodiode and micro lens are adopted for low noise, low smear and high sensitivity. This chip also features a strong anti-blooming and electronic shutter with variable charge-storage time.

Feature

- Micro Lens for high sensitivity
- Image-lag is negligible and excellent blooming suppression is performed.
- TTL level(5V) operation on HCCD & RG electrodes.
- 16 pin plastic-DIP.
- Variable electronic shutter of 1/50 to 1/100,000 sec.
- High sensitivity and low smear.

Pin Configuration

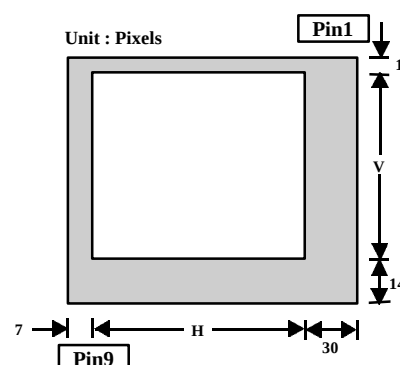


16 Pin Plastic - DIP
(Top View)

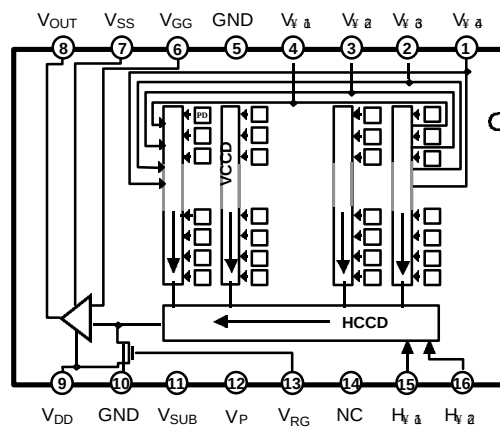
Structure

- Architecture : IT - CCD
- Optical size : 1/3 inch format
- Chip size : 6.0(H) x 5.2(V) $\pm$
- Number of effective pixels :
500 (H) x 582 (V) about 290K pixels
- Number of total pixels :
537 (H) x 597 (V) about 320K pixels
- Pixel size : 9.8 (H) x 6.3 (V) μ m²
- Optical black area
Horizontal direction : Front 7 pixels Rear 30 pixels
Vertical direction : Front 14 pixels Rear 1 pixels
- Number of dummy bits
Horizontal : 16
Vertical : 1 (Even field only)

Optical black position(Top View)



Block Diagram



Pin Description

No.	Symbol	Description	No.	Symbol	Description
1	$V_{V\ 4}$	Vertical register transfer clock 4	9	V_{DD}	Output amplifier drain bias
2	$V_{V\ 3}$	Vertical register transfer clock 3	10	GND	Ground
3	$V_{V\ 2}$	Vertical register transfer clock 2	11	V_{SUB}	Substrate(Overflow drain)bias
4	$V_{V\ 1}$	Vertical register transfer clock 1	12	V_P	Protection bias
5	GND	Ground	13	V_{RG}	Reset gate clock
6	V_{GG}	Output amplifier gate bias	14	NC	No connection
7	V_{SS}	Output amplifier source bias	15	$H_{V\ 1}$	Horizontal register transfer clock 1
8	V_{OUT}	CCD Output signal	16	$H_{V\ 2}$	Horizontal register transfer clock 2

Absolute Maximum Ratings

Parameter	Symbol	Value	Unit
Substrate voltage	$V_{SUB} - GND$	-0.3 to +55	V
Supply voltage	$V_{DD}, V_{OUT}, V_{SS} - GND$	-0.3 to +18	V
	$V_{DD}, V_{OUT}, V_{SS} - V_{SUB}$	-55 to +10	V
Vertical clock input voltage	$V_{V\ 1}, 2, 3, 4 - GND$	-10 to +20	V
	$V_{V\ 1}, 2, 3, 4 - V_P$	-0.3 to +27	V
	$V_{V\ 1}, 2, 3, 4 - V_{SUB}$	-55 to +10	V
Horizontal clock input voltage	$H_{V\ 1}, H_{V\ 2} - GND$	-10 to +15	V
Between vertical clock input pins	$V_{V\ i} - V_{V\ j}$	-10 to +15	V
Between horizontal clock and vertical clock input pins	$H_{V\ i}, H_{V\ j} - V_{V\ k}$	-17 to +17	V
Output pin voltage	$V_{RG}, V_{GG} - GND$	-10 to +15	V
	$V_{RG}, V_{GG} - V_{SUB}$	-55 to +10	V
Protective circuit voltage	$V_P - V_{SUB}$	-65 to 0.3	V
Storage temperature	T_{STG}	-30 to 80	°C
Operation temperature	T_{OPR}	-10 to 60	°C

Bias Condition

Parameter	Symbol	Min.	Typ.	Max.	Unit	Remark
Output amplifier drain voltage	V_{dd}	14.5	15.0	15.5	V	
Output amplifier gate voltage	V_{gg}	1.5	2.0	2.5	V	
Output amplifier source voltage	V_{ss}	Ground through 680Ω Resistor, ±5%				
Substrate voltage adjustment range	V_{sub}	5		15	V	
Fluctuation range after substrate voltage adjustment	ΔV_{sub}	-1		1	V	
Reset gate clock voltage adjustment range	V_{rgl}	0		4	V	
Fluctuation range after reset gate voltage adjustment	ΔV_{rgl}	-3		3	%	
Protection bias	V_p	Set to low level of vertical transfer clock				

DC Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Output amplifier drain current	I_{DD}	-	3	-	μA

Driving Condition

Parameter	Symbol	Min.	Typ.	Max.	Unit
Vertical clock high voltage	V_{H1}, V_{H3}	14.5	15.0	15.5	V
Vertical clock middle voltage	$V_{M1, 2, 3, 4}$	-0.2	0.0	0.2	V
Vertical clock low voltage	$V_{L1, 2, 3, 4}$	-9.0	-8.5	-8.0	V
Horizontal clock high voltage	$V_{H1, 2}$	4.5	5.0	5.5	V
Horizontal clock low voltage	$V_{L1, 2}$	-0.5	0.0	0.5	V
RG clock voltage difference	ΔV_{RGHL}	4.7	5.0	5.3	V
Substrate clock voltage	V_{SUB}	23	24	25	V

Electro-optical Performance

Item	Symbol	Min.	Typ.	Max.	Unit	Measurement Method	Remark
Sensitivity	SENS	55	70		μLux	1	
Saturation signal	V_{SAT}	800			μV	2	Temp=60; É
Smear	S_{MR}			0.015	%	3	
Blooming	BL			1	%	4	
Video signal shading	OSNU			15	%	5	
Dark signal level	V_{DARK}			2	μV	6	Temp=60; É
Dark signal shading	DSNU			2	μV	7	Temp=60; É

Measurement Method

1. Sensitivity

- Set to SILC (Standard Illumination Conditions*)
- Measure the average value of signal output (V_{out})
- Calculate the efficiency of V_{out} to light intensity

2. V_{sat}

- Adjust light intensity to 200 times of SILC
- Measure the average value of signal output

3. Smear

- Adjust light intensity to 200 times of SILC & readout clock
- Measure the signal output at horizontal optical black (V_{hopb})
- Measure the signal output at vertical blanking dummy (V_{vbd})
- $Smear = \{ (V_{vbd} - V_{hopb}) / V_{sat} \} \times 100 (\%)$

4. Blooming

- Adjust light intensity to 200 times of SILC & readout clock
- Measure the signal output at horizontal optical black (V_{hopb})
- Measure the signal output at blooming dummy area (V_{bd})
- $Blooming = \{ (V_{bd} - V_{hopb}) / V_{sat} \} \times 100 (\%)$

5. OSNU

- Set to SILC
- Measure the average value of signal output (V_{out})
- Measure the maximum value and the minimum value of signal output
- $OSNU = (V_{max} - V_{min}) / V_{out} \times 100 (\%)$

6. V_{dark}

- Measure the average value of signal output at dark condition

7. DSNU

- Measure the voltage difference between minimum and maximum of dark signal

* Standard Illumination Conditions

- Measure the average value of output of linear region
 - At this time, measure the light intensity of illumination at CCD face plate
 - Define SILC with above
 - Light source: Tungsten lamp(3100K)
 - Use a standard test lens at F8
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Notes on Handling

1) Static charge prevention

CCD image sensors are easily damaged by static discharge. Before handling be sure to take the following protective measures.

- a) Either handle bare handed or use non-chargeable gloves, clothes or material. Also use conductive shoes.
- b) When handling directly use an earth band.
- c) Install a conductive mat on the floor or working table to prevent the generation of static electricity.
- d) Ionized air is recommended for discharge when handling CCD image sensor.
- e) For the shipment of mounted substrates, use boxes treated for the prevention of static charges.

2) Soldering

- a) Make sure the package temperature does not exceed 80 °C.
- b) Solder dipping in a mounting furnace causes damage to the glass and other defects. Use a ground 30W soldering iron and solder each pin in less than 2 seconds. For repairs and remount, cool sufficiently.
- c) To dismount an image sensor, do not use a solder suction equipment. When using an electric desoldering tool, use a thermal controller of the zero cross On/Off type and connect it to ground.

3) Dust and Dirt protection

- a) Perform all assembly operations in a clean room (class 1000 or less).
- b) Do not either touch glass plates by hand or have any object come in contact with glass surfaces. Should dirt stick to a glass surface, blow it off with an air blower. (For dirt stuck through static electricity ionized air is recommended.)
- c) Clean with a cotton bud and ethyl alcohol if the grease stained. Be careful not to scratch the glass.
- d) Keep in a case to protect from dust and dirt. To prevent dew condensation, preheat or precool when moving to a room with great temperature differences.

4) Others

- a) Do not expose to strong light (sun rays) for long periods.
- b) Exposure to high temperature or humidity will affect the characteristics. Accordingly avoid storage or usage in such conditions.

Application Circuit

